

Teaching Exam

**Reasoning
Clock**

Lecture No.- 01



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Recap of Previous Lecture



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

Missing Number

Topics to be Covered



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

CLOCK



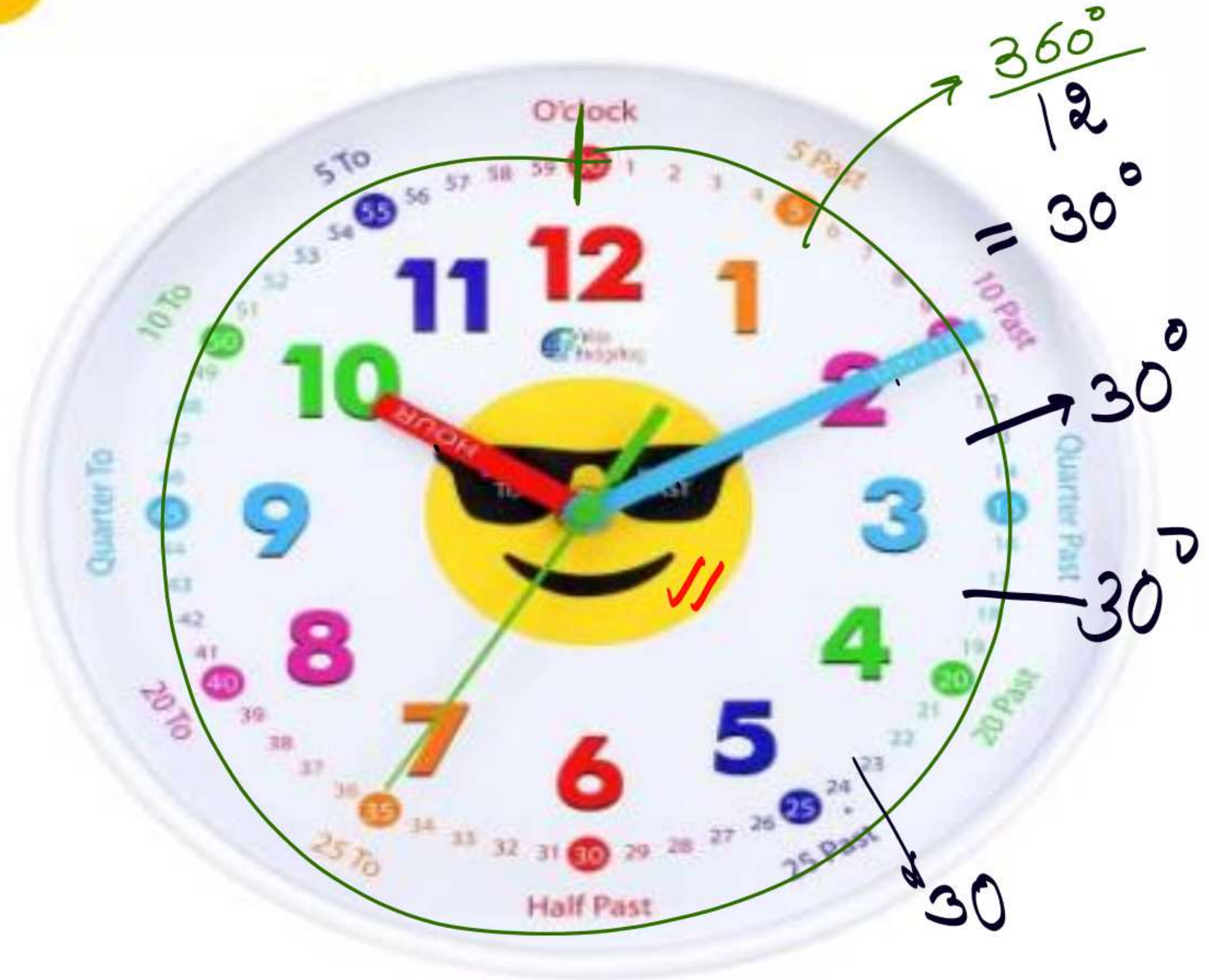
Question



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3 Types Hands

- ↓
- ⊙ Hour Hand
- ⊙ Minute Hand
- ⊙ Second Hand





Question

$$\begin{array}{l} \text{M.H} \\ \downarrow \\ 1 \text{ min} \rightarrow 6^\circ \end{array} \quad \left. \begin{array}{l} \text{H.H} \\ \downarrow \\ 1 \text{ min} = \frac{1}{2}^\circ \end{array} \right\}$$

$$\begin{array}{l} \text{S.H} \\ \downarrow \\ 1 \text{ sec} \rightarrow 6^\circ \end{array}$$



Minute Hand

$$\begin{array}{l} 1 \text{ Hour} = 360^\circ \\ [1 \text{ min} = 6^\circ] \end{array}$$

Hour Hand

$$\begin{array}{l} 1 \text{ Hour} = 30^\circ \\ [1 \text{ min} = \frac{1}{2}^\circ] \end{array}$$

Second Hand

$$\begin{array}{l} 1 \text{ min} = 360^\circ \\ 1 \text{ sec} = 6^\circ \end{array}$$



Question

Types

01 Finding the angle according to the given time.

02 Finding the time according to angle

03 Mirror Image of clock

04 Puzzle based questions

05 Miscellaneous questions

Mirror / water



$$\theta = 30H - \frac{11}{2} m$$

$$\text{Time} = \frac{(30H \pm \theta) \times 2}{11}$$



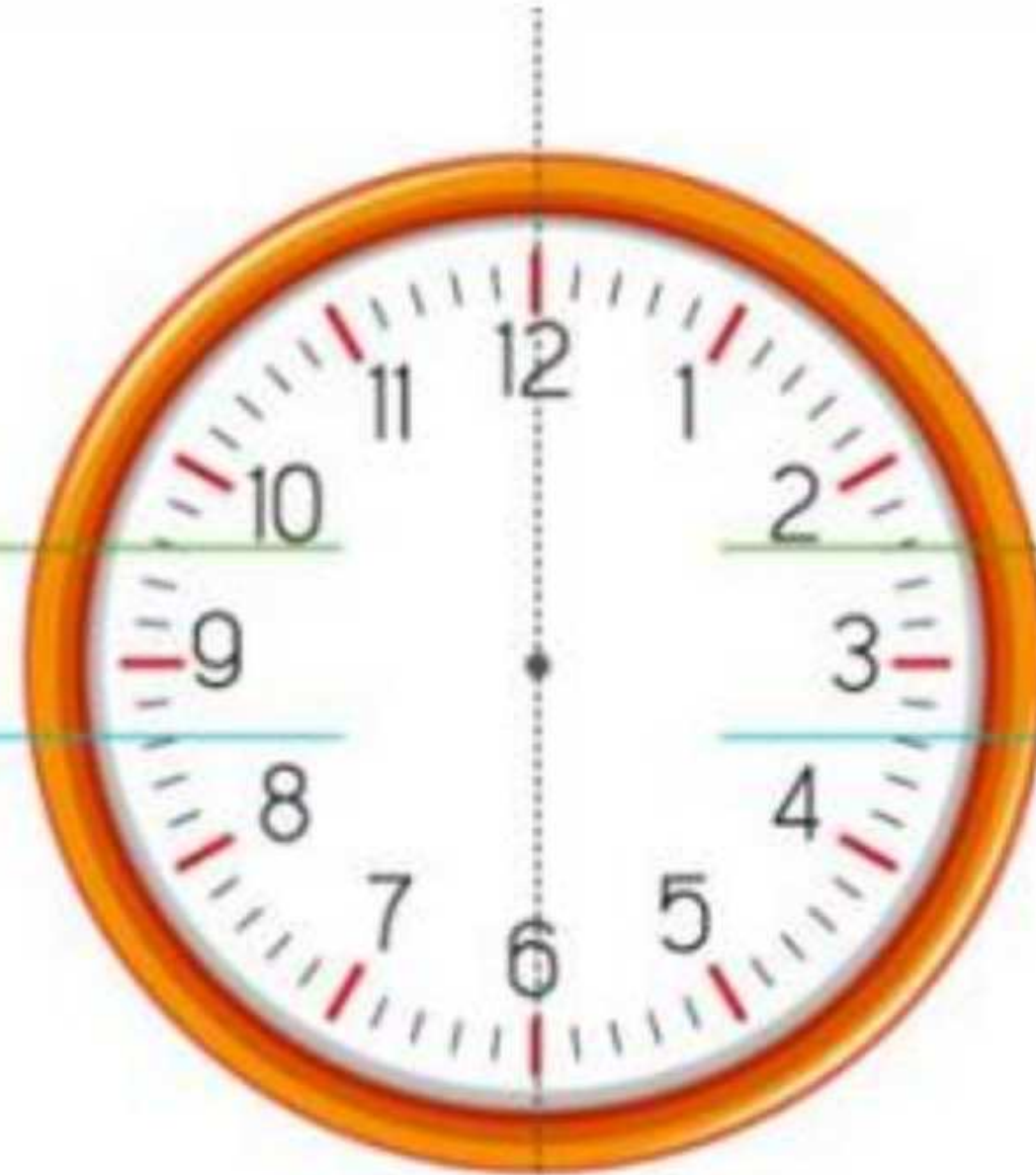
Question

Ist → Angle (θ) = ?
give → Time

$$\theta = 30H - \frac{11}{2}m$$

Angle →
→ छोटा [आन्तरिक] → 180° कम
→ बड़ा [बाह्य] → 180° से ज्यादा

2nd Half

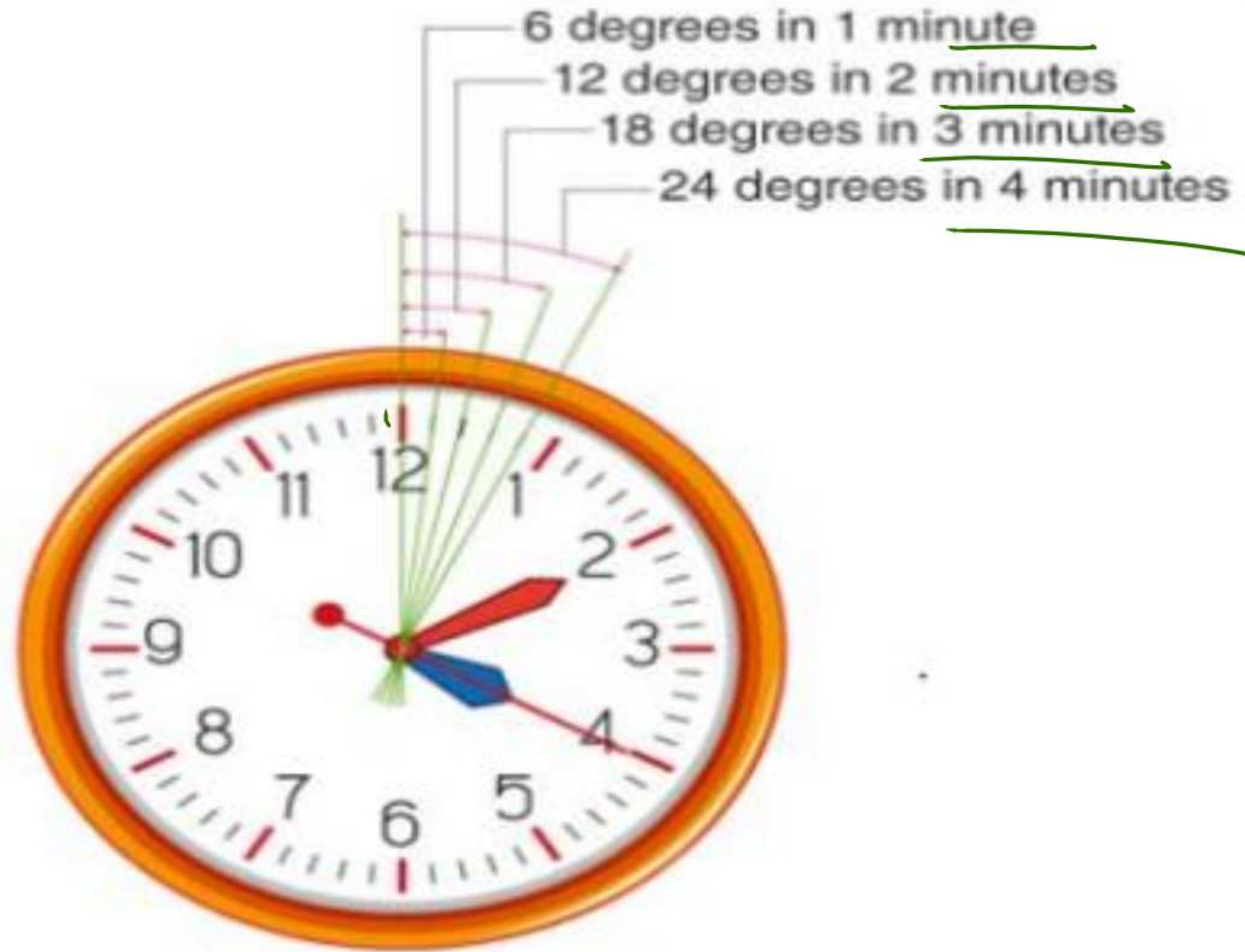


1st Half

+ve



Question





Question

Qo → घड़ी से $6:30^m$ बजे हैं, तो दोनो सुईयों के बीच कितना कोण बनेगा?

$$\begin{aligned} \text{A} \quad \theta &= 30H - \frac{11}{2}m \\ &\quad \downarrow \quad \quad \downarrow \\ &\quad 6 \quad \quad 30 \\ \theta &= 30 \times 6 - \frac{11}{2} \times 30 \\ &= 180 - 165 \\ &= 15^\circ \end{aligned}$$

$$\theta = 30H - \frac{11}{2}m$$

↓
Hour

↓
minutes

दिए गए समय का Ratio $1:5/1:6$ है तो मिनट का आधा कोण बनेगा





Question

10:10 Am [Angle]

Q. →

सुबह 10:10 बजे घंटे व मिनट की

सूई के बीच बनने वाला छोटा कोण

क्या होगा?

Ans

$$\theta = 30H - \frac{11}{2}M$$

$$\theta = 30 \times 10 - \frac{11}{2} \times 10$$

$$= 300 - 55$$

$$= 245^\circ$$

$$\begin{array}{r} 360 \\ - 245 \\ \hline 115^\circ \end{array}$$





Question

#Q. 45 सेकण्ड में सेकण्ड की सूई द्वारा बनाया गया कोण क्या होगा?

What will be the angle made by the second needle in 45 seconds?

(a) 300°

(b) 290°

(c) 270°

(d) 280°

45 sec

Sec Hand

↓
 $1 \text{ min} = 360^\circ$

$1 \text{ sec} = 6^\circ$

$45 \times 1 \text{ sec} \rightarrow 6^\circ \times 45$

$45 \text{ sec} \rightarrow \underline{270^\circ}$



Question

Important Facts

सम्पाती

Coincide (0 degree)

12 hrs $\rightarrow$ 11 times

24 hrs $\rightarrow$ 22 times

विपरीत कोण

Opposite angle (180 degrees)

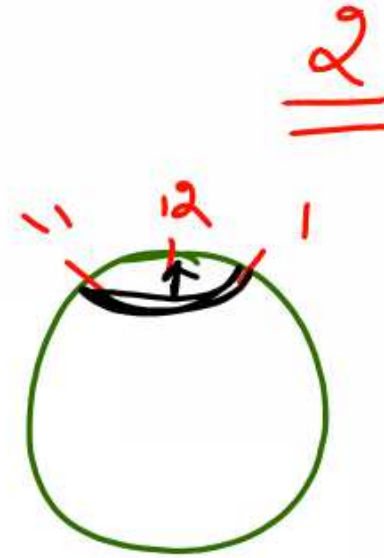
12 hrs $\rightarrow$ 11 times

24 hrs $\rightarrow$ 22 times

Right angle (90 degrees)

12 hrs $\rightarrow$ 22 times

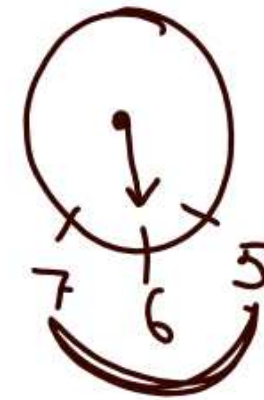
24 hrs $\rightarrow$ 44 times



Time = ?

given $\rightarrow$ θ [Angle]

$0^\circ / 180^\circ / 90^\circ$



$\downarrow$ 1 hr $\rightarrow$ 2912

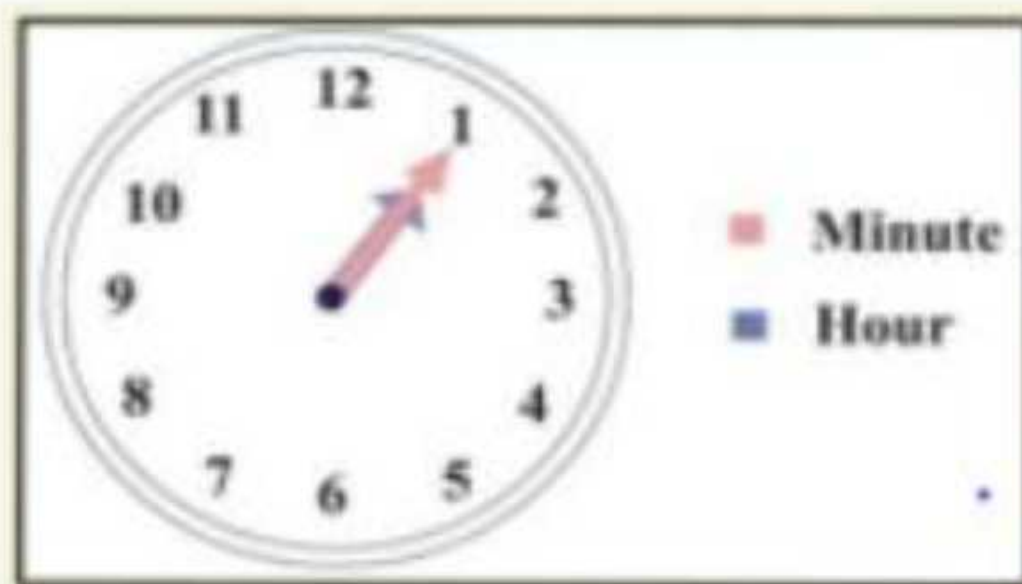


Question



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Coincide (Zero Degrees)



Qo → 3-4 વર્ગે ને વીચ જાણ 0° ના
જોણ વમેગા ?

3 - 4

$$\text{Time} = (30H \pm \theta) \frac{2}{11}$$

$\downarrow$ First $\downarrow$ Angle

A $(30 \times 3 \pm 0^\circ) \frac{2}{11}$
 $(90 \pm 0^\circ) \frac{2}{11} \Rightarrow \frac{180}{11} \Rightarrow 16 \frac{4}{11}$ [મિનટ]
3:16 $\frac{4}{11}$ વર્ગે //



Question



Opposite (180 Degrees)

Approx 1 hr

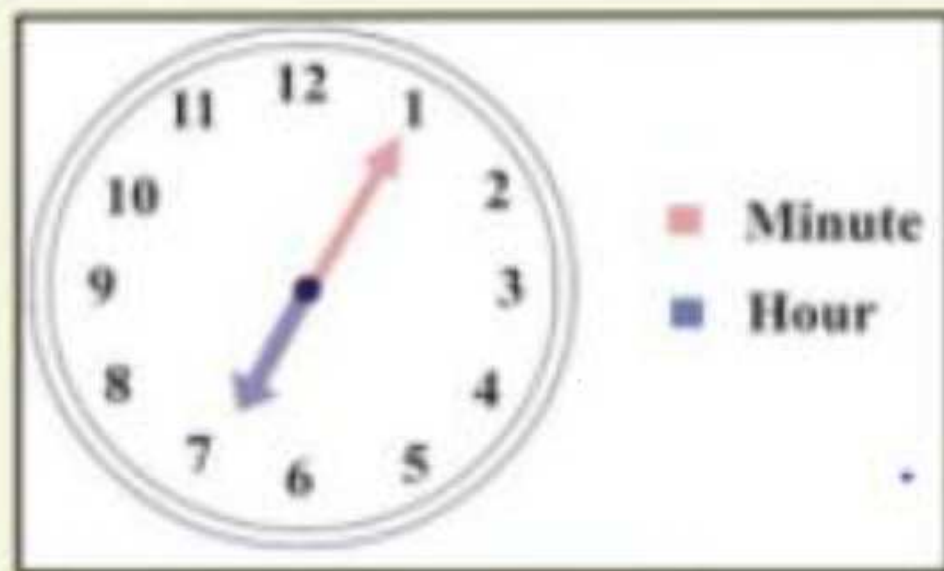
1 time

***11 *11**

i.e, 12 hrs -> 11

times 24 hrs -> 22

times



Q → 7-8 बजे के बीच कब घड़ी की दोनों सुईया 30 min की दूरी / विपरीत होगी ?

A

30 min
 $\downarrow$
 $\times 6$
 $\underline{180^\circ \rightarrow 0}$

H → 7 }
 0 → 180° }

Time = $(30 \times 7 \pm 180) \frac{2}{11}$

Time = $(30 \times 7 - 180) \frac{2}{11}$
 $= (210 - 180) \frac{2}{11}$
 $= \frac{60}{11} \Rightarrow 5 \frac{5}{11} \text{ min}$
 7: 5 $\frac{5}{11}$ बजे



Question



Right Angle (90 Degrees)

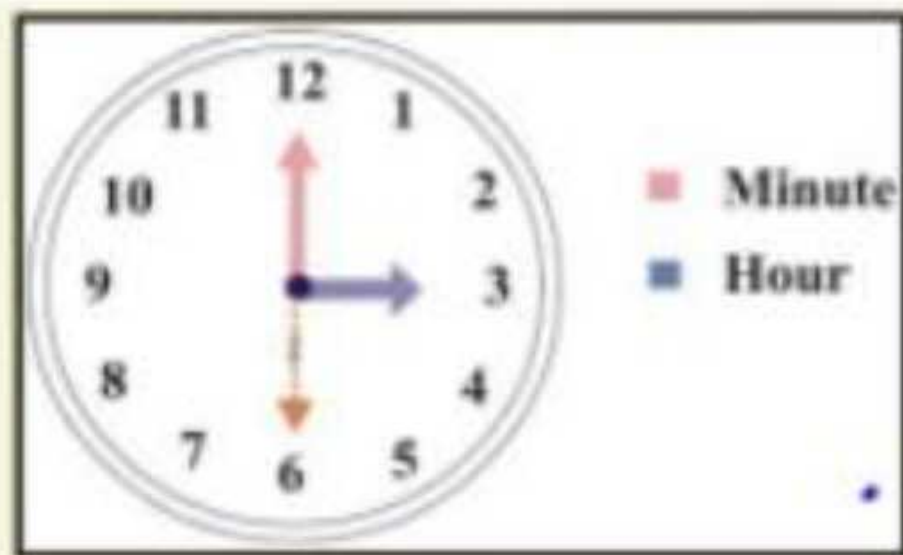
Approx 1 hr

2 times

i.e, 12 hrs 22

times 24 hrs $\rightarrow$ 44

times



Q. 1-2 बजे के बीच कब
दोनों सुईया समकोण बनाएंगी?

$$\text{Time} = \frac{(30H \pm \theta)}{11} \quad \swarrow \quad 90^\circ [\text{R.A}]$$

$$\begin{aligned} H &\rightarrow 1 \\ \theta &= 90^\circ \end{aligned}$$

$$\begin{aligned} (30H \pm \theta) &\rightarrow (30 \times 1 \pm 90) \\ (30 + 90) &\quad (30 - 90) \\ \frac{120}{11} &\quad \frac{-60}{11} \\ 10 \frac{10}{11} &\quad 5 \frac{5}{11} \text{ min} \\ 1:10 \frac{10}{11} &, 1:21 \frac{5}{11} \end{aligned}$$



Question

Image

Water Image

Mirror Image

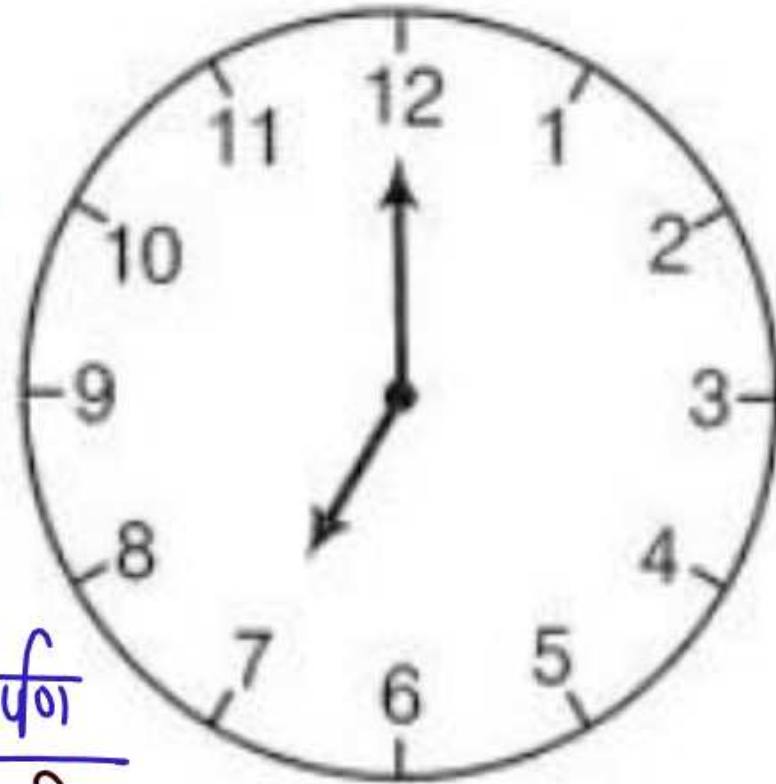
18:30 / 17:30

- ઘડી / જલ

જલ / ઘડી

11:60
- ઘડી / દર્પણ

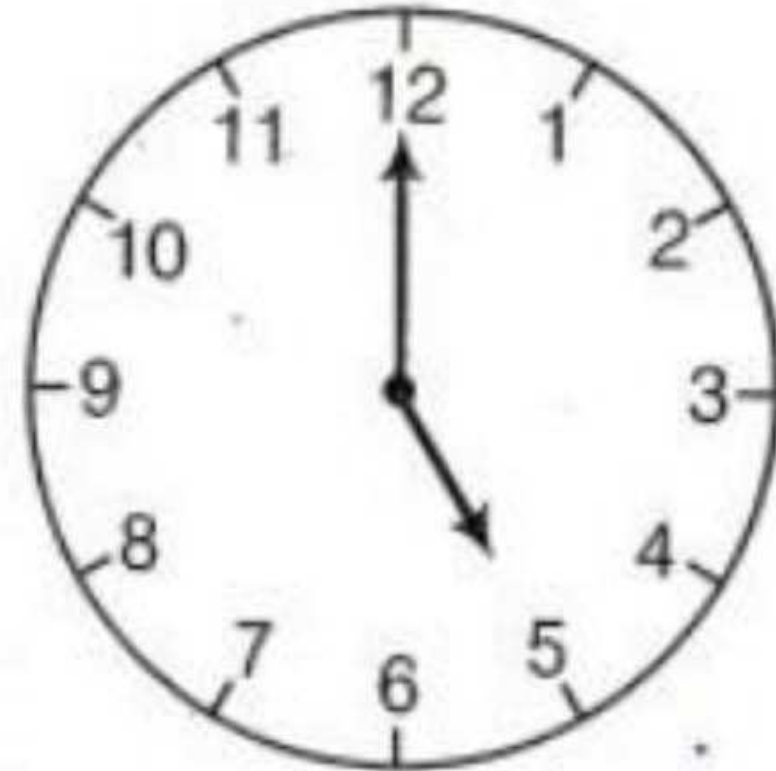
દર્પણ / ઘડી



ઘડી



લમ્બવત્ દર્પણ



દર્પણ પ્રતિબિમ્બ



Question

#Q. यदि किसी घड़ी में 4 बजकर 30 मिनट हो रहे हैं, तो इस स्थिति में दर्पण प्रतिबिम्बित समय क्या होगा?

If a clock is showing 4:30, then what will be the mirror reflected time in this case?

(a) 7:32 $\rightarrow$ 7:32

(b) 8:30 $\rightarrow$ 8:30

☒ (c) 7:30 $\rightarrow$ 7:30

(d) 7:39 $\rightarrow$ 7:39

$$\begin{array}{r} 11:60 \\ - 4:30 \\ \hline 7:30 \end{array}$$



2 mins Summary



TEACHING
WALLAH

Topic

One

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Four

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Five

CLOCK

THANK YOU